

Exercises - Bayes' Theorem

1. Company A supplies 40% of the computers sold and is late 5% of the time. Company B supplies 30% of the computers sold and is late 3% of the time. Company C supplies another 30% and is late 2.5% of the time. A computer arrives late - what is the probability that it came from Company A?
2. In Orange County, 51% of the adults are males. One adult is randomly selected for a survey involving credit card usage. It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration). Use this additional information to find the probability that the selected subject is a male.
3. A person uses his car 30% of the time, walks 30% of the time and rides the bus 40% of the time as he goes to work. He is late 10% of the time when he walks; he is late 3% of the time when he drives; and he is late 7% of the time he takes the bus.
 - a. What is the probability he took the bus if he was late?
 - b. What is the probability he walked if he is on time?
4. In a study of pleas and prison sentences, it is found that 45% of the subjects studied were sent to prison. Among those sent to prison, 40% chose to plead guilty. Among those not sent to prison, 55% chose to plead guilty.
 - a. If one of the study subjects is randomly selected, find the probability of getting someone who was not sent to prison.
 - b. If a study subject is randomly selected and it is then found that the subject entered a guilty plea, find the probability that this person was not sent to prison.
5. On a game show, a contestant can select one of four boxes. The red box contains one \$100 bill and nine \$1 bills. A green box contains two \$100 bills and eight \$1 bills. A blue box contains three \$100 bills and seven \$1 bills. A yellow box contains five \$100 bills and five \$1 bills. The contestant selects a box at random and selects a bill from the box at random. If a \$100 bill is selected, find the probability that it came from the yellow box.

6. A plane's "black-box" is manufactured by only 3 companies: AirCorp, BigSkies, and CharterUS - who make 80%, 15%, and 5% of all the black-boxes made, respectively. Invariably, some of these are defective. Assuming the percentage of defective black-boxes made by AirCorp, BigSkies, and CharterUS are 4%, 6%, and 9%, respectively, find the probability that a randomly selected black-box from all black-boxes made that is found to be defective came from AirCorp.
7. Consider 3 coins where two are fair, yielding heads with probability 0.50, while the third yields heads with probability 0.75. If one randomly selects one of the coins and tosses it 3 times, yielding 3 heads - what is the probability this is the biased coin?
8. Assume the probability of having tuberculosis (TB) is 0.0005, and a test for TB is 99% accurate. What is the probability one has TB if one tests positive for the disease?
9. An automobile manufacturer has three factories: A, B, and C. They produce 50%, 30%, and 20% respectively, of a specific model of car. 30% of the cars produced in factory A are white, 40% of those produced in factory B are white, and 25% produced in factory C are white. If an automobile produced by the company is selected at random, find the probability that it is white. Given that an automobile selected at random is white, find the probability that it came from factory B.
10. Two manufacturers supply blankets to emergency relief organizations. Manufacturer A supplies 3000 blankets and 4% are irregular in workmanship. Manufacturer B supplies 2400 blankets and 7% are found to be irregular. Given that a blanket is irregular, find the probability that it came from manufacturer B.